BRIAN YILDIRIM

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EDUCATION

University of California, Irvine - B.S. in Computer Science — Expected Jun. 2027 | Irvine, CA

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Systems Engineering, Software Testing, Computer Networks, Statistical Data Analysis, Discrete Math, Calculus II, Linear Algebra

PROJECTS

StrivoAI: Career Intelligence Platform | *TypeScript, React, Supabase, PostgreSQL, Docker, Stripe* — Oct. 2025 - Jun. 2026

- Cut resume-analysis wait time by redesigning a sequential AI pipeline into parallel model calls with schema validation, caching and retry fallbacks.
- Shipped five AI career workflows across **41** Supabase Edge Functions with authentication, recoverable background jobs and replay-safe Stripe billing.
- Blocked regressions before merge with a **7**-stage pull-request gate (lint, types, Vitest, Deno, database, Playwright, build) running **2,800+** tests.

Minecraft Enchantment Calculator | *Python, Flask, JavaScript, Tailwind CSS* — May 2025 - Aug. 2025

- Built a memoized recursive optimizer modeling Minecraft anvil rules, finding the cheapest enchantment order within the game's **39**-level limit.
- Launched it as a public web app with SEO-ready pages, earning **2.8M+** Google Search Console impressions at an **8%** click-through rate.

Rate Limiter Library | *C++, C API, CMake, GoogleTest, Google Benchmark* — Aug. 2025 - Sep. 2025

- Built a thread-safe C++20 rate limiter with lock-free lookups over a sharded token-bucket table, scaling to **1M** keys at **~134** bytes per key.
- Found flooded single keys made every core fight over one cache line; added a read-only denial path so denials scale across threads.
- Verified correctness with **54** tests in Ubuntu/macOS CI plus ASan and TSan builds, and fixed an eviction test failing **6–28** of **1,000** runs.

Image Quality Gate | *Python, FastAPI, OpenCV, NumPy, Docker* — May 2025 - Jun. 2025

- Built a FastAPI microservice that rejects blurry or badly exposed uploads, returning the measurements behind each decision so callers see why.
- Added a diagnostics endpoint and paired before/after benchmarks, confirming no measurable slowdown on the default /quality path.
- Shipped it production-style with Docker, Prometheus metrics, request IDs and GitHub Actions CI running lint, type checks and **130** tests on every push.

Concurrent Key-Value Server | *Python, TCP Sockets, Multithreading, File I/O* — Mar. 2025 - Apr. 2025

- Built a multithreaded TCP key-value server with a custom SET/GET/REMOVE/PRINT protocol, capped at **100** concurrent clients by default.
- Prevented lost updates by guarding shared state with a reentrant lock and persisting every change, so data survives restarts; covered by **54** tests.

ACTIVITIES

UCI ICS Student Council Hackathon | *Python, Django, Supabase, Docker* — Oct. 2025 - Nov. 2025

- Led a team of engineers to ship a campus activities map in under a week, building its Django REST API on Supabase.
- Designed indexed Supabase tables to speed up event queries while the API supported 500+ concurrent users.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL/PostgreSQL
Backend & Data: Django, FastAPI, Node.js, Supabase, NumPy
Frontend & UI: React, Tailwind CSS
DevOps & Testing: Docker, GitHub Actions, CMake, GoogleTest, Prometheus